Daedalus POC — Gap Analysis

Generated from comparing the POC document (28 sections) against all implemented code across 5 projects.

Legend

- ✅ **Complete** — Fully implemented and tested
- 🟡 **Partial** — Implemented but incomplete or missing features
- 🔴 **Missing** — Not implemented at all
- ⬜ **Out of Scope** — Explicitly deferred per POC (Phase 2+)

1. Python Library (daedalus_watch_kit)

POC Section	Requirement	Status	Notes
§11 Library Architecture	Package structure, tests	✅	12 modules, 10 test files, 141 tests passing
§12 Stage 1: Ingestion	Load IMU, HR, ECG, events, annotations	✅	Full schema validation, completeness report
§12 Stage 2: Time Sync	Offset, drift, cross-correlation	✅	Offset estimation, linear drift, sync error computation
§12 Stage 3: Windowing	6 window configs with overlap	✅	All 6 configs implemented (SQI, HRV, tremor, gait, stress)
§12 Stage 4: Signal Quality	HR_SQI, IMU_SQI, bad-segment detection	✅	From HR and from PPG variants, classify_bad_segments
§13 HRV: IBI Cleaning	Ectopic beat removal, interpolation	✅	clean_ibi with physiological range + step-change filters
§13 HRV: Time Domain	mean_hr, sdnn, rmssd, pnn50, nn_outlier_rate	✅	All 5 features
§13 HRV: Frequency Domain	lf_power, hf_power, lf_hf_ratio, total_power	✅	Welch + Lomb-Scargle methods
§13 HRV: Nonlinear	sample_entropy	🟡	sample_entropy implemented; dfa_alpha1 and hrv_triangular_index **missing**
§13 HRV: PPG pipeline	Bandpass, baseline removal, motion cancellation	🔴	Not implemented — assumes no raw PPG access (per POC assumption)
§14 Stress: Personalized Baseline	5-min rest baseline metrics	✅	PersonalBaseline dataclass

POC Section	Requirement	Status	Notes
§14 Stress: Rule-based Score	z-score formula, 0-100 normalization	✅	compute_stress_score with configurable weights
§14 Stress: Classifier	Logistic regression, RF, XGBoost	🟡	PersonalizedStressClassifier with threshold-based binary classification; **ML models not implemented**
§14 Stress: Evaluation	F1, false alarm rate, leave-one-out	🟡	evaluate_stress_detection returns F1/precision/recall/false_alarm; **cross-validation not implemented**
§15 Tremor: Features	band_power, dominant_freq, spectral_entropy, etc.	✅	All 7 features computed
§15 Tremor: Classification	Rule-based + probability output	✅	Both rule-based and probabilistic classification
§15 Tremor: STFT	2-sec window, 50Hz, FFT 128/256, 50% overlap	✅	Via scipy.signal.stft
§16 Gait: Detection	Peak detection, autocorrelation	✅	Both methods implemented
§16 Gait: Cadence	60/dominant_period, 4-sec windows	✅	Autocorrelation-based + peak-counting fallback
§16 Gait: Regularity	cadence_variability, harmonic_ratio, gait_bout_length	✅	All 3 features
§16 Gait: Freezing Proxy	Arm swing reduction, variability, context	✅	detect_gait_instability_candidate
§17 Haptic: Patterns	Gait metronome, breathing, tremor grounding	✅	All 3 pattern generators + latency measurement
§17 Haptic: Metrics	Latency, jitter, missed pulse	✅	compute_haptic_performance with all targets

POC Section	Requirement	Status	Notes
§18 Validation Report	Per-session HTML report	🔴	**Not implemented** — no report generation (Bland-Altman, scatter plots, confusion matrices)
§21 Data Quality Rules	Session rejection, window flagging	✅	All thresholds in quality_rules.py
§9 File Schema	Parquet/CSV/JSON schemas	✅	schemas.py + matching CSV writers in apps

2. Wear OS Watch App (daedalus-watch)

POC Section	Requirement	Status	Notes
§5 Watch App	Wear OS data-collection app	✅	Full app with foreground service
§5 Start/Stop recording		✅	Via SensorCollectionService
§5 Sensor status check		✅	getSensorStatus() debug display
§5 Battery logging		🟡	Battery level available via BatteryManager but **not actively logged**
§5 Local storage		✅	CSV files in /data/raw/{subject}/{session}/
§5 Periodic sync to phone		🟡	Wear OS DataLayer (Android) + BLE Central (iPhone) — **DataLayer receives but doesn't auto-push**
§5 Event markers		✅	Sync taps, sync claps buttons
§5 Haptic cue engine		✅	HapticEngine with all 3 patterns
§5 Debug screen		✅	Status display with IMU availability, collecting state, vibrator
§5 IMU at 50Hz		✅	ImuSensorManager at 20000µs (50Hz)
§5 HR continuous		✅	Health Services API via callbackFlow
§5 HRV if exposed		🟡	Attempted via Health Services; HRV not confirmed available
§5 Raw PPG		⬜	Per POC assumption — not expected to be available

POC Section	Requirement	Status	Notes
§5 GPS		⬜	Not implemented (medium priority, disabled by default)
§5 Haptic event timestamps		✅	Logged to watch_haptics.csv
§5 Foreground service 60min		✅	SensorCollectionService with foreground notification
§10 Time Sync: Triple tap		✅	SYNC_TAPS event logged
§10 Time Sync: Five claps		✅	SYNC_CLAPS event logged
§7 Block 0: Consent/setup		🔴	**No UI for participant metadata collection**
§7 Block 2: Rest baseline		🟡	Event button exists; **no protocol timer/guidance**
§7 Block 3: Paced breathing		🟡	Haptic breathing pattern exists; **no guided breathing UI**
§7 Block 4: Stress task		🟡	Event button exists; **no stress task UI or score collection**
§7 Block 5: Movement matrix		🔴	**No movement task UI** (10 movement types)
§7 Block 6: Tremor tasks		🔴	**No tremor task UI** (5 tremor tasks)
§7 Block 7: Gait protocol		🔴	**No gait task UI** (7 gait tasks)
§7 Block 8: Haptic tests		✅	Haptic test button + patterns available
§7 Block 9: Recovery/questionnaires		🔴	**No post-session questionnaire UI**
§22 Privacy: Participant IDs only		✅	Uses subject_id/session_id, no names
§22 Regulatory language		✅	No treatment claims in app text

3. Android Companion App (daedalus-companion)

POC Section	Requirement	Status	Notes
§5 Phone App	Companion Android app	✅	Full app with session management
§5 Pair with watch		✅	Wear OS DataLayer (WatchDataListener + WatchDataClient)
§5 Receive watch streams		✅	DataLayer receives IMU/HR/events
§5 Record ECG/chest strap		✅	ChestStrapBLEManager parses RR intervals from Cycplus H2
§5 Record event labels		✅	Event marker buttons + auto-logged events
§5 Record session metadata		✅	SessionMetadata → session_metadata.json
§5 Store files locally		✅	CSV files in /Android/data/.../daedalus/data/raw/
§5 Export session package		✅	Zip export with all session files
§5 Live signal quality debug		🟡	HR display shows live data; **no signal quality visualization**
§10 NTP sync		⬜	Relies on Android system time (auto-NTP)
§7 Protocol blocks UI		🟡	3 event buttons (Rest, Stress, Sync); **missing 7 protocol block types**
§8 Event Label Dictionary		✅	All events defined in shared EventLabels
§9 File structure		✅	Matches daedalus_watch_kit schema exactly

POC Section	Requirement	Status	Notes
§21 Data quality: reject criteria		🔴	**No session validation/rejection logic**
§22 Privacy		✅	Participant IDs only, no names stored

4. iOS Companion App (daedalus-companion-ios)

POC Section	Requirement	Status	Notes
§5 Phone App	iPhone companion for watch	✅	SwiftUI app with dual BLE
BLE Peripheral (watch→iPhone)		✅	WatchBLEManager with custom Daedalus service
BLE Central (chest strap→iPhone)		✅	ChestStrapBLEManager with Heart Rate Service (0x180D)
Session management		✅	SessionManager with start/end/export
Event markers		✅	Rest, Stress, Sync buttons
Export to zip		✅	Zip via /usr/bin/zip, iOS share sheet
French localization		✅	Localizable.strings (fr.lproj)
Live signal display		✅	HR + RR from chest strap, watch HR
§7 Protocol blocks UI		🟡	3 event buttons; **missing full protocol flow**
§9 File schema match		✅	CSV/JSON output matches Python library

5. Cross-Cutting / Infrastructure

POC Section	Requirement	Status	Notes
§4 Hardware: Watch5 Pro		✅	minSdk 30, Wear OS 3+
§4 Hardware: Cycplus H2		✅	BLE Heart Rate Service parsing
§4 Hardware: Video recording		🔴	**No video recording integration** (for labeling)
§4 Hardware: Nice-to-haves		🔴	Foot pod, GAITRite, respiratory belt, metronome, BP cuff — **none implemented**
§6 Phase 0: Founder pilot (N=1)		🟡	Pipeline works; **no automated validation**
§6 Phase 1: Healthy cohort (N=10-20)		🔴	**No multi-subject management, no cohort diversity tracking**
§6 Phase 2: PD pilot		⬜	Out of scope per POC sequencing
§19 Validation Report		🔴	**No automated report generation** (Bland-Altman, scatter plots, confusion matrices, histograms)
§19 HR vs ECG validation		🔴	**No comparison logic built**
§19 HRV validation		🔴	**No correlation analysis built**
§19 Signal quality report		🟡	signal_quality_report() exists; **no HTML output**
§19 Haptic performance report		🟡	Metrics computed; **no visualization**

POC Section	Requirement	Status	Notes
§22 Data versioning (DVC)		🔴	**Not set up**
§22 Encrypted drives		⬜	Infrastructure concern, not app code
§23 Risk: Battery drain		🟡	Foreground service runs; **no battery monitoring/logging**
§23 Risk: Haptic latency		✅	Measured and logged; trigger-to-onset tracked
§24 Minimum Viable Dataset		🔴	**No dataset aggregation or summary tools**
§25 Investor deliverables		🔴	**No demo dashboard, no proof report generator**
§26 Day 1-3 checklist		✅	All 72-hour tasks covered by implemented code
§27 Primary metric: Signal quality		✅	HR_SQI and IMU_SQI fully implemented

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Summary by Priority

Critical Gaps (Block Phase 0 Validation)

1. 🔴 **Validation report generator** (§18) — No automated Bland-Altman plots, scatter plots, confusion matrices, or HTML reports
2. 🔴 **Protocol block UI** (§7 Blocks 2-9) — Only 3 of 10 protocol blocks have UI buttons; no guided flow, timers, or task instructions
3. 🔴 **Video recording integration** (§4) — No camera integration for labeling movement/gait/tremor events
4. 🔴 **Battery monitoring** (§5, §23) — Battery level not logged during sessions

Important Gaps (Block Phase 1 Scale-Up)

5. 🔴 **Multi-subject management** (§6) — No subject registration, consent tracking, or cohort diversity logging
6. 🔴 **Session validation/rejection** (§21) — No automated criteria checking (IMU completeness >95%, sync error <250ms, etc.)
7. 🔴 **HR vs ECG comparison** (§19) — No statistical comparison logic built into the pipeline
8. 🟡 **ML classifiers** (§14) — Only rule-based stress scoring; no logistic regression, RF, or XGBoost

Nice-to-Have Gaps

9. 🔴 **Nonlinear HRV features** (§13) — dfa_alpha1 and hrv_triangular_index missing
10. 🔴 **Post-session questionnaires** (§7 Block 9) — No stress/fatigue/comfort score collection UI
11. 🔴 **Data versioning (DVC)** (§22) — Not set up
12. 🔴 **Metronome app integration** (§4) — No external metronome for paced walking/breathing

Out of Scope (Deferred)

- PD pilot protocol (§20) — Requires clinical approval, out of scope for Phase 0/1
- Raw PPG pipeline (§13) — Per POC assumption, not available to third-party developers
- Freezing of Gait detection (§16) — Explicitly deferred ("too advanced for first proof")
- Speech task (§20) — PD pilot addition
- Investor deliverables (§25) — Post-Phase 1

Effort Estimate to Close Critical Gaps

Gap	Effort	Priority
Protocol block UI (watch + phone)	2-3 weeks	P0
Validation report generator (Python)	1-2 weeks	P0
Battery monitoring + logging	2-3 days	P0
Video recording integration	1 week	P1
Session validation/rejection	3-5 days	P1
HR vs ECG comparison logic	3-5 days	P1
ML classifiers (stress)	1-2 weeks	P2
Multi-subject management	1 week	P2

Total to close P0 gaps: ~4-6 weeks (aligns with POC's Week 2-5 timeline)